

CSCE 679 DATA VISUALIZATION · MODULE 11 · LECTURE NOTES

Scale

When there are more data points than pixels.

MODULE OUTCOMES

- Explain overplotting and its consequences.
- Explain binning and density as the general answer.
- Explain sampling and when it is acceptable.
- Explain the rendering and interaction constraints.
- Visualise a dataset larger than the display.

1 Overplotting

OVERPLOTING DESTROYS THE DENSITY INFORMATION, WHICH WAS USUALLY THE FINDING

When marks overlap, the figure saturates — a region containing a hundred points and a region containing ten thousand look identical — so the densest part of the data is exactly where the figure conveys least, which inverts what you wanted.

And it is order-dependent: the last points drawn are visible and the first are hidden beneath them, so the figure depends on the row order of the input file, which is arbitrary and frequently sorted by something meaningful — making the visible points a biased subset.

Which means the apparent outliers are the only reliable feature of an overplotted figure — they are precisely the points with nothing drawn on top of them — and **the main mass is uninterpretable**, which is the opposite of the usual intention.

So partial transparency is the first fix and it is not sufficient: it saturates too, at a higher threshold, and the threshold depends on whatever alpha value you happened to choose — which makes it another undisclosed parameter (§2's callout).

2 Binning

THE IDEA

there are more data points than pixels, so aggregate to the pixel grid and encode the count on colour. One mark per bin, not one per point.

WHICH GIVES

2D histogram / heatmap	rectangular bins
hexbin	hexagonal bins, with more uniform neighbour distances and less visual aliasing
datashader-style	one bin per pixel, with the count scaled carefully

AND THE SCALING OF THE COUNT MATTERS

linear colour hides the structure outside the densest region entirely
 log or a rank-based scale reveals it
 which is Module 08 section 3's binning choice again, with the same consequence

SO: the colour scale on a density plot is as consequential as the bin size, and both should be stated.

The colour scale on a density plot is as consequential as the bin size — a linear scale hides everything outside the mode, because the mode's count may be several orders of magnitude above the rest and compresses all the remaining structure into the bottom of the scale. **The count-scaling point is what distinguishes a good density plot from a dark smudge with a bright spot**, and it is the single most common defect in large-data figures.

AND BINNING IS A DERIVATION, SO THE BINS ARE AN ANALYTICAL CHOICE

The bin size changes the apparent structure — too fine and the figure shows noise; too coarse and the structure is averaged away — which is Module 05 §2's histogram problem in two dimensions and with the same lack of a principled answer.

And the bin boundaries can create or destroy apparent features, particularly **where the data has a natural grid or a rounding artefact** — values rounded to the nearest five will produce stripes at some bin sizes and not others, and the stripes are in the binning rather than in the world.

So report the bin size and the count scaling, and check the figure at two or three different bin sizes before trusting any feature you see in it — which takes minutes and is the equivalent of CSCE 676 Module 04 §3's stability check.

Which is the discipline this whole course repeats: any choice that changes the picture is an analytical decision and belongs in the caption (Module 13 §2), and this module adds four such choices to the list.

3 Sampling

- **For the overall shape, a sample is entirely sufficient — ten thousand points show the same distribution as ten million, and show it considerably more legibly** since they do not overplot.
- **And it is very much cheaper** — in rendering time, in network transfer, and in interaction latency (CSCE 671 Module 11 §3's thresholds, which this module has to meet).
- **But it loses the rare cases, and the rare cases are frequently the point — outliers, anomalies, and small subgroups all disappear from a uniform sample**, which is CSCE 676 Module 08's subject vanishing from the figure.
- **So: sample for the shape and keep the extremes — a stratified sample that deliberately over-represents the tails**, which is explicit, effective, and easy to implement.
- **And state the sampling**, because **a reader will assume they are seeing everything. An unstated sample is a figure about data the reader does not know was selected** — and if the sampling was non-uniform, the figure is about a distribution that does not exist.

4 The system constraints

- **Pixels.** A display has a few million of them, so **anything beyond that is aggregated whether you chose to aggregate or not — the only question is whether you chose the aggregation** or let the rendering order choose it for you (§1).
- **Transfer.** Sending a billion points to a browser is simply not possible, so **the aggregation has to happen server-side — which is why the architecture is the design** rather than an implementation detail (CSCE 676 Module 10).
- **Interaction latency.** Under 100 ms for a brush to feel direct (CSCE 671 Module 11 §3), **which bounds what can be recomputed on each interaction** and therefore bounds what interactions you can offer at all.
- **So: precompute the aggregations you will need, which is a data cube, and is how interactive large-data visualisation actually works** — the interaction becomes a lookup in a precomputed structure rather than a query over the raw data.
- **And degrade deliberately — show a sample while the user is dragging and the full aggregate on release**, which meets the latency budget where recomputation cannot. **Precomputing the aggregations turns an interactive query into a lookup, and is the only way the latency budget is met** at this scale.

AND THE HONEST SUMMARY OF THIS MODULE

At scale you are always showing a summary — binned, sampled, or aggregated, and frequently all three — **so the question is which summary and whether you said so.**

Which is CSCE 676 Module 01 §2's framing exactly: an approximation with a stated bound is a result, and an exact answer you cannot compute is not an answer — and the visualisation case adds that the approximation is visible and looks exact.

So state the aggregation method, the bin size, the count scaling, and any sampling — four things, and the figure is then assessable by somebody who did not make it.

And the risk is specific: a density plot looks authoritative and is a function of four choices the reader cannot see — which is the same problem as Module 08's map and has exactly the same fix, which is to state them.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The dashshader documentation and its principles (free)	§2 as a working system — and its discussion of count scaling is the best treatment of that specific problem.
Carr et al. — Scatterplot matrix techniques for large N (free)	§1 and §2 in the original — where hexagonal binning was introduced, for exactly this reason.
Liu, Jiang & Heer — imMens: Real-time Visual Querying of Big Data (free)	§4's data cube approach — precomputed aggregation for interactive latency.
Lins, Klosowski & Scheidegger — Nanocubes (free)	§4's structure — the aggregation index that makes the interaction a lookup.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Plot ten million points directly** and report what you can read.
- 2 Reverse the row order** and compare the two figures.
- 3 Add transparency** and find the alpha at which it saturates.
- 4 Hexbin the same data** and compare.
- 5 Render the density with linear and log count scaling**, and report what appears.

- 6 **Vary the bin size three ways** and check whether a feature survives.
- 7 **Find a rounding artefact** that appears at one bin size and not another.
- 8 **Sample uniformly and check whether a known outlier survives.**
- 9 **Build a stratified sample** that keeps the tails, and compare.
- 10 **Measure your interaction latency** against the 100 ms threshold.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does overplotting destroy exactly the information you wanted?
- 2 Why is it order-dependent, and what follows?
- 3 Why is transparency insufficient?
- 4 Give three binning approaches and what each offers.
- 5 Why does the count scaling matter so much?
- 6 Why is binning an analytical choice, and what should you check?
- 7 What does a sample preserve and what does it lose?
- 8 What is the fix, and why must sampling be stated?
- 9 Name four system constraints and the architectural consequence.
- 10 State the module's summary and the four things to report.